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Enterprise SaaS Retention Analysis: Data Engineering and Analytics Report for Keeplo

Prepared by: Leonardo Castellanos - Analytics Division

Project Scope: End-to-End Pipeline - Business Intelligence

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Chapter 1: Executive Summary

This strategic briefing provides a comprehensive evaluation of the subscription retention architecture developed for Keeplo, establishing a data verified baseline to guide executive decision making. By auditing the transactional ledger through a rigorous analytics pipeline, this report translates raw operational database logs into high level business intelligence to address critical customer attrition points.

The analytical framework rests on the core institutional mandate that protecting recurring revenue from the existing subscriber base is fundamentally more profitable than expanding front end marketing acquisition funnels. The following sections isolate the core financial health indicators, dissect systemic user lifecycle friction, and document the data engineering infrastructure required to secure sustained enterprise growth.

1.1 Core Financial and Operational Diagnostics

The baseline corporate metrics for Keeplo indicate a stabilized Monthly Recurring Revenue of \$95,285 sustained by an active subscriber base of 1,789 users. While these indicators demonstrate solid market traction, corporate profitability is heavily constrained by a monthly Churn Rate of 27.5%, which accounts for a net loss of 568 customers within the current reporting cycle. This severe volume leak indicates that while customer acquisition pipelines are successfully converting initial demand, the platform faces an immediate utility bottleneck that compounds customer lifetime value degradation.

The underlying revenue ledger proves that capital generation remains heavily dependent on the premium subscription tier which accounts for 72% of total contract volume, leaving the remaining 28% distributed across basic contracts. Consequently, stabilizing retention within the high value premium segment represents the single most critical operational priority to mitigate compounding monthly revenue contraction.

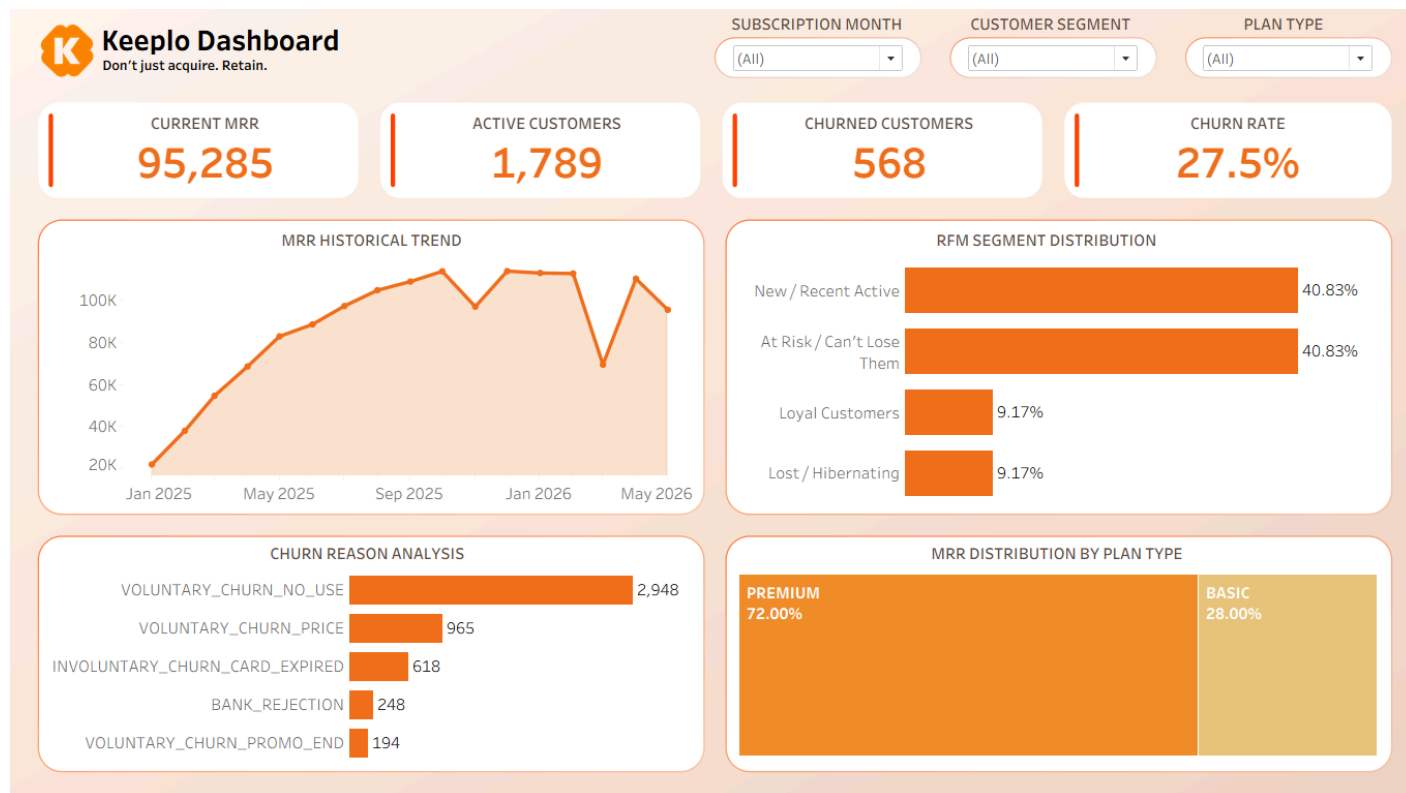


Figure 1: Keeplo Corporate Retention Dashboard displaying core performance indicators, historical monthly recurring revenue trends, and plan composition analytics.

1.2 Systemic Revenue Leakage Analysis

An evaluation of behavioral customer archetypes exposes a critical risk distribution within the user lifecycle matrix. The Recency, Frequency, and Monetary calculation model isolates a matching mathematical spread where 40.83% of total unique accounts are classified as New or Recent Active users, while an identical 40.83% have already deteriorated into the At Risk category. This equilibrium exposes a severe structural breakdown immediately following the initial conversion funnel, where onboarding mechanisms fail to transition new signups into sustained platform adoption.

This operational deficit is validated by the transactional logs, which identify voluntary product abandonment due to a lack of core feature usage as the primary threat to revenue stability. This specific churn category accounts for 2,948 customer cancellations, completely eclipsing price sensitive attrition which accounts for 965 cancellations, proving that customer loss is driven by onboarding failure rather than market pricing pressures.

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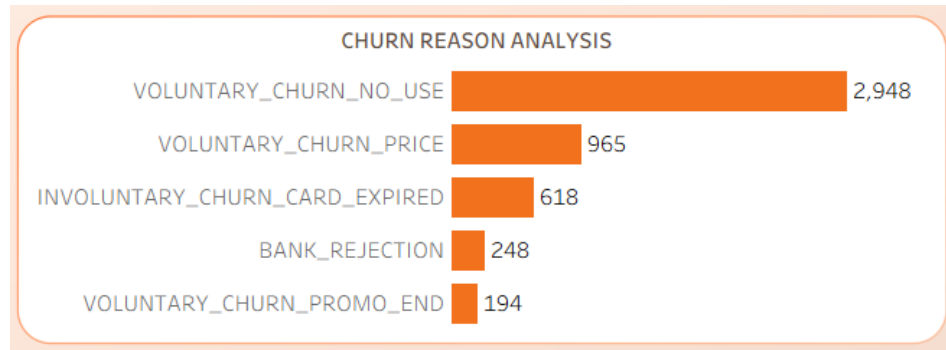


Figure 2: Quantitative breakdown of historical cancellation triggers isolating voluntary product abandonment as the primary threat.

1.3 Data Engineering Integrity and Mitigation Blueprint

To guarantee absolute operational clarity for leadership, the insights presented in this report originate from a fully audited data infrastructure layer designed to eliminate data pipeline noise and transactional anomalies. The production dbt pipeline successfully countered critical upstream quality deficits, executing the deletion of duplicate account records, the removal of negative accounting errors, and the normalization of fragmented contract fields that previously obscured true revenue distribution.

With data integrity established, reversing the 27.5% churn threshold requires an immediate two pronged operational strategy. The product team must deploy automated feature activation protocols targeted at the 40.83% At Risk user pool to stimulate engagement, while the engineering team must integrate automated dunning management into the billing gateway to neutralize credit card expirations and bank rejections, which currently account for a combined total of 866 preventable customer cancellations.

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Chapter 2: Introduction and Objectives of the Project

This chapter establishes the foundational groundwork for the Keeplo business intelligence initiative, outlining the strategic necessity and structural goals that driven the implementation of this analytics framework. In the rapidly evolving software as a service landscape, corporate sustainability depends heavily on maximizing customer lifetime value and minimizing churn dynamics. This documentation sets the parameters for an intervention designed to move the company away from anecdotal reporting toward a highly systematic, data driven governance model. By aligning data infrastructure with high level commercial priorities, this project provides executives with the visibility required to protect recurring revenue streams and optimize long term operational stability.

2.1 Strategic Context and Business Mandate

The foundational doctrine anchoring this project is summarized by the corporate directive to prioritize customer retention over aggressive front end acquisition funnels. While traditional growth strategies place an asymmetric focus on customer acquisition metrics, the financial realities of a subscription model prove that scaling relies on minimizing user attrition. High customer acquisition costs require multiple months of uninterrupted subscription tenure just to achieve capital break even on each account. Consequently, a monthly churn rate of 27.5% represents an immediate erosion of marketing efficiency and long term profitability.

This analysis moves beyond aggregate figures to unpack the exact friction points within the user lifecycle, empowering leadership to transform raw behavioral signals into proactive preservation campaigns.

2.2 Analytical Scope and Solution Objectives

The core technical and strategic objective of this business intelligence deployment is to establish an uncompromised, real time diagnostic layer over the entire Keeplo user base. This solution maps customer journeys from initial signup through historical milestone cycles, translating disparate transactional logs and session event tables into an integrated behavioral matrix. Operatively, the system isolates high risk cohorts before complete attrition occurs, allowing retention teams to deploy targeted interventions based on specific usage deficits.

By institutionalizing automated metrics like Recency, Frequency, and Monetary scoring, the platform removes subjectivity from customer health evaluations. Ultimately, the analytical architecture serves as a strategic decision engine, providing executive leadership with the

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precise visibility needed to execute data driven risk mitigation, optimize product onboarding protocols, and secure the overarching subscription landscape.



Chapter 3: Data Architecture and Information Engineering

This chapter details the data engineering framework and information architecture deployed to establish a single source of truth for Keeplo. Raw transactional records and behavioral user logs are inherently chaotic, containing structural anomalies and reporting noise that can distort high level executive metrics if left unremediated. To prevent strategic misalignment, a robust data pipeline was constructed to ingest, clean, and model data from core operational databases into highly optimized analytics assets. The following sections document the structural mapping of the core data sources and the systematic data cleansing protocols implemented to guarantee absolute data integrity across all corporate performance dashboards.

3.1 Data Lineage and Structural Source Mapping

The data architecture relies on three primary data pipelines that capture distinct dimensions of the customer relationship: account demographics, ledger transactions, and platform interaction logs. The customer core repository establishes the master profile data, mapping critical baseline attributes such as signup dates, acquisition channels, and contract tier types. This foundational layer intersects directly with the transactional ledger database, which logs monthly historical billings, credit card authorization attempts, and explicit cancellation reason codes.

Finally, the user activity system captures granular interaction events, monitoring features accessed, session durations, and support ticket volumes. Through an analytics engineering layer, these independent operational sources are systematically compiled into two unified star schema models, isolating structural lifecycle behavior on one side and long term historical revenue milestones on the other.



Figure 3: Overview of the data model.

3.2 Data Quality Remediation and Operational Cleansing

Before provisioning data to the executive visibility layer, the engineering pipeline executed an extensive auditing protocol to eliminate systemic data corruption and anomalies embedded within the raw data streams. Initial data audits exposed critical quality deficits, including duplicated customer identifiers that artificially inflated subscriber counts, contorted date formats from legacy entry systems, and negative billing amounts that introduced severe corporate revenue miscalculations.

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Furthermore, user interaction tables contained extreme session duration outliers and chronic chronological incoherencies, while the plan classification column suffered from fragmented capitalization variables that split single plan types into multiple artificial products. The deployment of automated cleansing protocols successfully neutralized these threats by enforcing strict text uniformization via uppercase casting, purging corrupted negative values, isolating extreme activity outliers, and deduplicating customer profiles. This technical remediation layer guarantees that the active subscriber figures and monthly recurring revenue numbers presented to leadership are free from operational distortion and completely validated for corporate governance.



Chapter 4: Methodology and Technical Implementation in Tableau

This chapter outlines the methodological framework and analytical configurations executed within Tableau to transform the cleaned data infrastructure into an interactive corporate intelligence engine. Developing a corporate grade dashboard requires moving beyond default software configurations to engineer customized data synchronization and advanced calculated dimensions. To achieve this, the implementation strategy focused on establishing unified global filtering mechanics and deploying advanced calculation layers that operate independently of worksheet dimensions. The following sections detail the synchronization protocols that ensure seamless dashboard interactivity and the deployment of advanced Level of Detail expressions to isolate precise user lifecycle metrics.

4.1 Global Filter Synchronization and Interactive Architecture

The interactive architecture of the dashboard is designed to provide executive stakeholders with an intuitive drill down experience without compromising visual alignment or data consistency. In a multi source data environment, default analytical filters operate in isolation, which can lead to fragmented views and conflicting metrics across different charts. To eliminate this operational risk, the dashboard utilizes customized global scoping mechanics, ensuring that primary filters such as Subscription Month, Customer Segment, and Plan Type are structurally tied across independent data repositories.

This synchronization guarantees that when a director isolates a specific customer segment, the entire analytical canvas, including historical Monthly Recurring Revenue trends, core key performance indicators, and plan compositions, recalculates instantly. Furthermore, by locking all elements into absolute floating container coordinates with zero padding, the interface maintains its premium aesthetic integrity during high volume query execution.



Figure 4: Filter section showing an example set.

4.2 Advanced Analytical Engineering via Level of Detail Expressions

To capture the true operational reality of the customer lifecycle, the analytics layer integrates advanced Level of Detail expressions utilizing the FIXED syntax. Standard reporting tools calculate metrics based purely on what is visually displayed on the chart grid, which prevents the simultaneous evaluation of individual customer records against macro corporate baselines. The integration of advanced calculations overcomes this



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limitation by anchoring specific metrics directly to the unique customer identifier, completely independent of the visualization shelf dimensions.

This approach allowed for the successful deployment of the Recency, Frequency, and Monetary matrix, dividing users into precise behavioral blocks based on their true historical activity. Additionally, this methodology enabled the engineering of a dual axis hybrid trend component, fusing high density line graphics with translucent area shading to deliver premium interface aesthetics while accurately reflecting historical capital momentum.

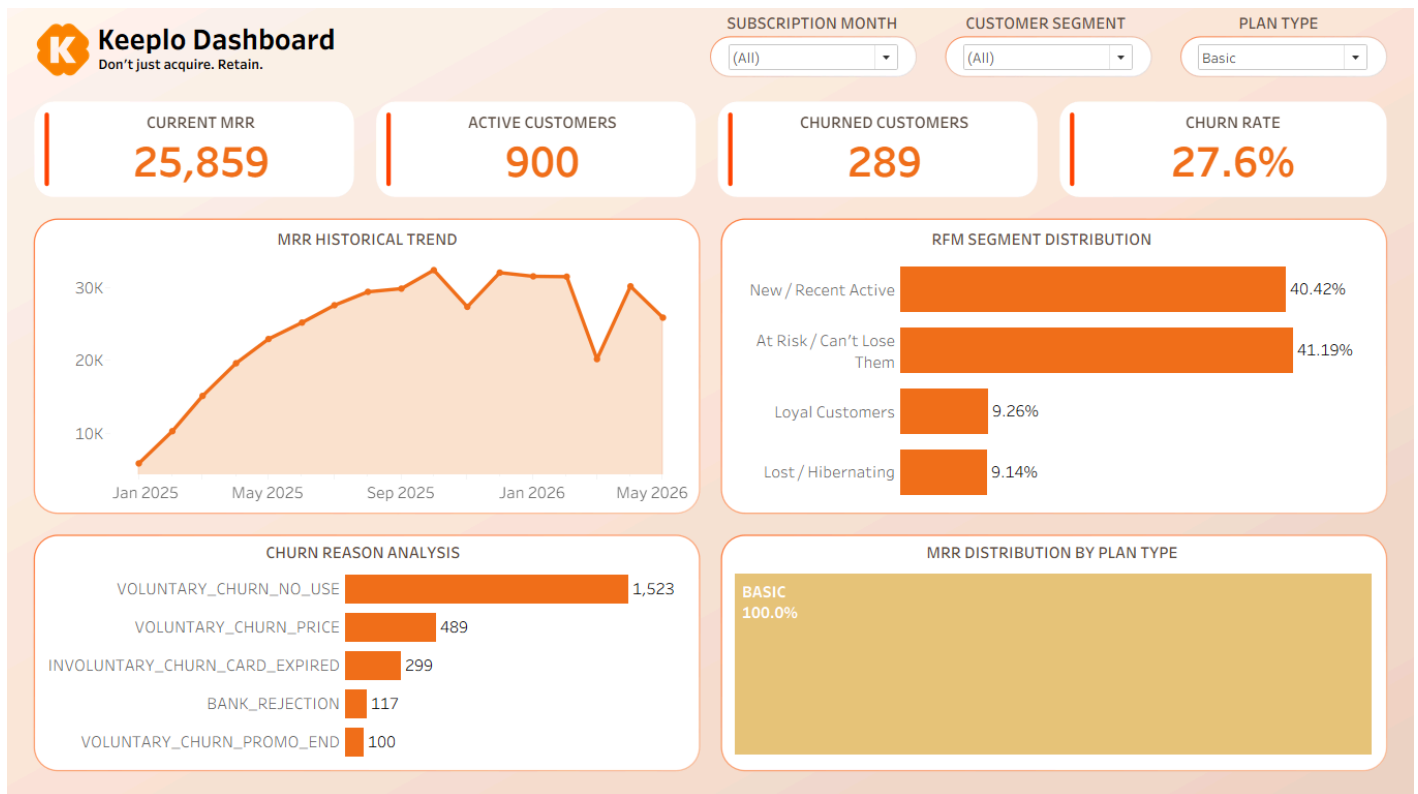


Figure 5: Close-up of the global filtering interface demonstrating multi-dimensional synchronization across independent subscription metrics.



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Chapter 5: Analysis of Findings and Analytical Diagnostics

This chapter delivers a rigorous empirical diagnostic of the commercial performance data captured across the Keeplo operating environment. Moving beyond superficial metrics, this evaluation cross references macroeconomic revenue timelines with granular user behavioral distributions to expose the root causes of subscriber attrition. By evaluating the structural relationship between contract categories, platform interaction frequencies, and explicit transaction failure records, leadership can isolate the precise operational vulnerabilities threatening corporate growth. The following evaluations provide the empirical evidence required to transition from reactive troubleshooting toward targeted, high impact customer preservation strategies.

5.1 Historical Revenue Momentum and Timeline Trends

A chronological evaluation of the Monthly Recurring Revenue ledger from January 2025 through May 2026 reveals a distinct trajectory characterized by a prolonged accumulation phase followed by an acute contraction corridor. The platform experienced sustained upward momentum throughout the first three quarters of 2025, driven by successful customer acquisition funnels and compounding contract expansions that pushed revenue to historical peaks. However, this growth phase was abruptly interrupted by a sharp contractive correction in early 2026, exposed by a severe drop in monthly billings before stabilizing at the current baseline of \$95,285 sustained by 1,789 active users.

This stabilization valley indicates that the business has reached a critical structural equilibrium where front end acquisition volumes are merely replacing churned capital rather than expanding the net revenue footprint, underscoring the urgent necessity of stabilizing the existing customer foundation.

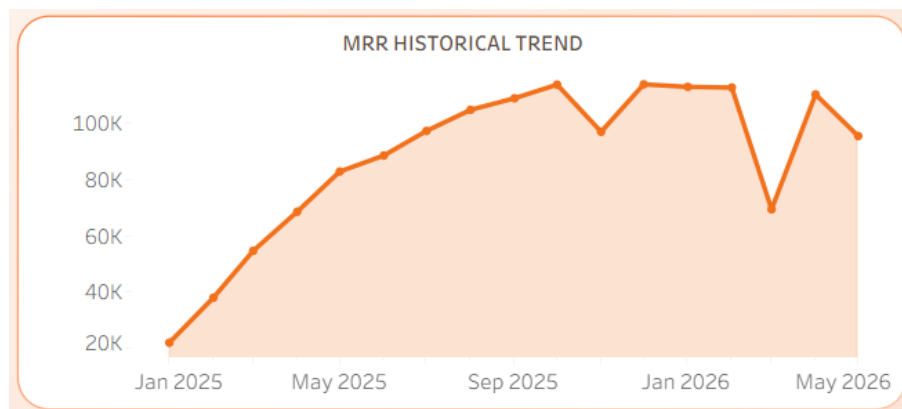


Figure 6: Historical Monthly Recurring Revenue trajectory from January 2025 through May 2026, isolating growth deceleration phases and current revenue stabilization.



5.2 Recency, Frequency, and Monetary Behavioral Distributions

The application of the Recency, Frequency, and Monetary scoring matrix exposes a highly polarized customer engagement topology across the active subscriber base. The analytical data demonstrates a symmetric distribution where 40.83% of total accounts reside in the New or Recent Active segment, while an identical 40.83% are locked within the At Risk category, leaving the remaining volume marginally distributed between Loyal and Lost cohorts. This mathematical distribution points to a systemic lifecycle friction point, demonstrating that customers are experiencing a severe drop in platform interaction immediately after completing their initial onboarding cycle.

Users are failing to transition into the high engagement frequencies that characterize loyal customers, creating a highly volatile risk concentration where nearly half of the active customer portfolio is highly susceptible to imminent cancellation.

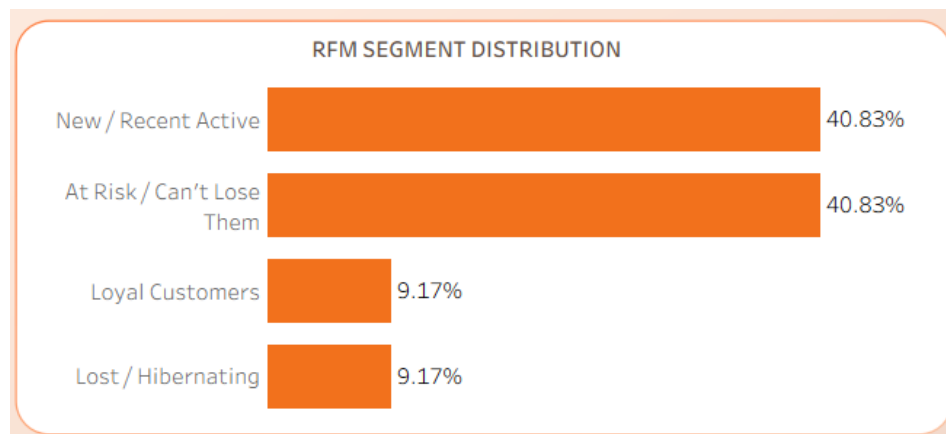


Figure 7: Historical Recency, Frequency, and Monetary scoring matrix from January 2025 through May 2026.

5.3 Granular Churn Triggers and Contract Portfolio Vulnerabilities

A deep dive into explicit cancellation indicators confirms that product utility deficits represent a far greater threat to capital retention than competitive pricing pressures. The transactional ledger isolates voluntary churn driven by a lack of platform usage as the single largest driver of revenue loss, accounting for 2,948 absolute customer cancellations, whereas price sensitivity triggered only 965 cancellations. This loss pattern is heavily compounded by portfolio vulnerabilities tied to plan distribution, where high value premium contracts account for 72% of total revenue compared to the 28% generated by basic tiers. Because corporate financial stability is disproportionately reliant on the premium segment, voluntary attrition within these high value tiers exponentially accelerates revenue degradation.



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Finally, this revenue leakage is aggravated by transactional friction within the payment processing gateway, where preventable involuntary triggers including credit card expirations and bank rejections account for 866 customer losses, representing a severe operational leakage that requires immediate infrastructural remediation.

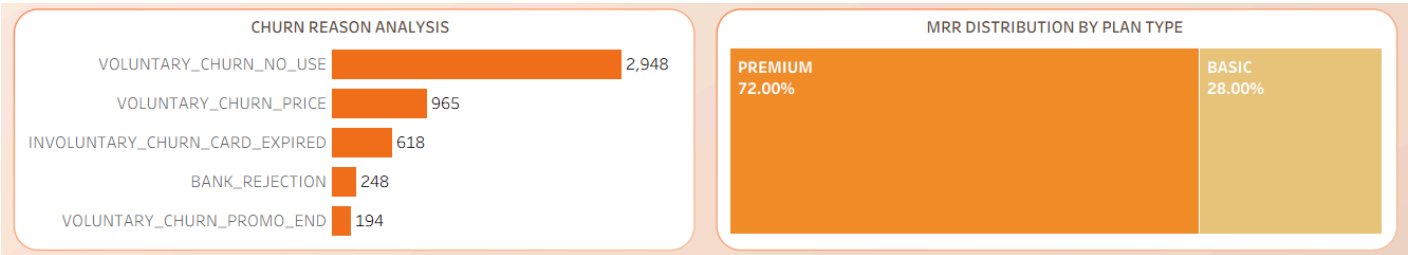


Figure 8: Historical Churn Reason Analysis and MRR Distribution by Plan Type from January 2025 through May 2026.



Chapter 6: Strategic Recommendations and Action Plan

This chapter converts the empirical findings compiled in the diagnostic phases into an actionable operational roadmap designed to defend Keeplo recurring revenue. To effectively reverse the current churn trajectory, management must deploy targeted interventions that directly address the specific failure modes exposed within the customer behavior ledger. By aligning product engineering, customer success workflows, and billing operations, the organization can systematically mitigate both voluntary and involuntary attrition vectors. The following subsections deliver specific, high impact strategic initiatives tailored to stabilize the existing subscriber base, protect high value segments, and maximize long term corporate asset value.

6.1 Mitigation of Voluntary Attrition via Targeted Feature Engagement

Addressing the primary driver of subscription loss requires an immediate transformation of the post conversion user journey, specifically targeting the 2,948 cancellations caused by a lack of platform utility. Because 40.83% of the total user portfolio resides within the critical At Risk category, the customer success division must abandon reactive support methodologies in favor of automated, data driven engagement protocols. The product team should implement real time monitoring triggers that detect a deceleration in core feature access within the first thirty days of subscription tenure. Upon detection of low usage anomalies, automated lifecycle marketing workflows must deliver personalized, context specific feature discovery guides to demonstrate core value and accelerate product adoption.

By systematically reengaging this high risk pool before the subscription cycle ends, the organization can break the utility bottleneck and successfully guide users into sustained behavioral adoption.

6.2 Infrastructural Gateway Optimization for Involuntary Churn Remediation

To neutralize the 866 customer losses triggered by credit card expirations and bank rejections, the engineering division must execute a comprehensive optimization of the payment processing infrastructure. Involuntary churn represents an unnecessary loss of acquired capital that can be efficiently mitigated through automated transactional workflows. The business should integrate an automated dunning management framework into the primary billing gateway, substituting blunt collection attempts with intelligent retry logic that schedules transaction retries based on optimal bank authorization windows. Furthermore, the implementation of an automated account updater service will allow the billing platform to securely synchronize expired card credentials with major financial

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institutions before a payment rejection occurs. Resolving this technical friction layer protects existing revenue pipelines without requiring direct customer intervention, immediately driving down the macro churn baseline.

6.3 Premium Contract Protection and Retention Frameworks

Given that the premium subscription tier represents a disproportionate 72% of the aggregate Monthly Recurring Revenue, safeguarding this specific segment constitutes the single highest priority for financial governance. Management must establish a dedicated premium account retention framework, decoupling high value enterprise contracts from standard consumer workflows. Customer success teams must be assigned directly to top tier accounts, conducting proactive quarterly operational reviews to ensure maximum return on investment and address platform friction early. Additionally, the marketing and product structures should introduce tiered value incentives, such as priority feature access and advanced reporting capabilities, to reinforce the competitive advantage of remaining on the premium tier. Protecting the premium contract footprint insulates the financial core of the business, stabilizing revenue momentum while lower tier basic contracts are optimized.



Chapter 7: Conclusions and Future Work

This final chapter synthesizes the strategic conclusions derived from the Keeplo business intelligence deployment and outlines the scalability horizons for future engineering phases. Establishing a mature data infrastructure marks a critical pivot from historical data compilation toward active, predictive business governance. By resolving upstream quality deficits and providing an uncompromised visibility layer, this initiative arms leadership with the empirical foundation required to defend recurring revenue. The following sections provide a final balance of the governance evolution and define the subsequent operational milestones needed to secure extended market expansion.

7.1 Final Balance and Governance Evolution

The implementation of this retention analytics framework marks a definitive evolution in how Keeplo manages its core commercial assets. Prior to this intervention, operational blind spots and fragmented data layers masked the true extent of customer attrition, limiting executive interventions to reactive, short term troubleshooting. The unified dashboard addresses this deficit directly, providing a verified baseline of \$95,285 in Monthly Recurring Revenue sustained by 1,789 active users. While the 27.5% churn rate remains a significant operational challenge, the business is now equipped with the precise diagnostic transparency needed to implement targeted structural reforms. Transitioning to a data driven governance model ensures that strategic choices regarding product onboarding, pricing architecture, and infrastructure optimization are anchored in uncompromised empirical truth.

7.2 Future Work and Scalability Horizons

Moving into subsequent operational cycles, the data infrastructure must expand to accommodate predictive capabilities and automated workflow execution. The next technical phase will focus on migrating the production data pipeline into a fully automated cloud architecture, eliminating manual extraction steps and enabling real time data ingestion. Furthermore, engineering efforts should prioritize the integration of predictive machine learning algorithms directly into the data modeling layer to calculate individual customer churn probabilities before users enter the At Risk classification. Expanding the data lake to incorporate web analytics, detailed customer acquisition costs, and geographical market vectors will also enrich behavioral profiles.

By evolving the business intelligence solution from a diagnostic dashboard into a predictive decision system, Keeplo will successfully defend its subscription footprint and drive scalable, long term corporate growth.



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